



Simulation of pore space production law and capacity expansion mechanism of underground gas storage



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Abstract: One-dimensional gas injection storage building and one-cycle injection-production modeling experiment, and two-dimensional flat core storage building and multi-cycle injection-production modeling experiment were carried out using one-dimensional long core and large two-dimensional flat physical models to find out the effects of reservoir physical properties and injection-production balance time on reservoir pore utilization efficiency, effective reservoir capacity formation and capacity-reaching cycle. The results show that reservoir physical properties and formation water saturation are the main factors affecting the construction and operation of gas-reservoir type underground gas storage. During the construction and operation of gas-reservoir type gas storage, the reservoir space can be divided into three types of working zones: high efficiency, low efficiency and ineffective ones. The higher the reservoir permeability, the higher the pore utilization efficiency is, the smaller the ineffective working zone is, or there is no ineffective working zone; the smaller the loss of injected gas is, and the higher the utilization rate of pores is. The better the reservoir physical properties, the larger the reservoir space and the larger the final gas storage capacity is. The higher the water saturation of the reservoir, the more the gas loss during gas storage capacity building and operation is. Optimizing injection-production regime to discharge water and reduce water saturation is an effective way to reduce gas loss in gas storage. In the process of multiple cycles of injection and production, there is a reasonable injection-production balance time, further extending the injection-production balance period after reaching the reasonable time has little contribution to the expansion of gas storage capacity.

Key words: gas reservoir-type underground gas storage; multi-cycle injection and production; injection-production equilibrium time; pore utilization efficiency; effective gas storage volume

Introduction

Building and operation of underground gas storages (UGSs) are characterized by periodicity, high flow rate, quick injection-production and alternating load, in order to meet the needs for peak shaving in extreme cold weather and emergency responses [1–3]. In China, the existing UGSs are mainly reconstructed from gas reservoirs which have entered their middle and late development stages. In the process of UGS capacity construction, fluid displacement by gas is mainly conducted, in conjunction with gas huff-and-puff [4–6]. Influenced by factors such as

water invasion after formation pressure depletion, the distribution law of underground fluid becomes very complex [7], making it difficult to build UGSs.

Scholars have studied the influencing factors on key indices such as effective storage capacity and working gas volume of UGSs through indoor physical experiments. Tooseh et al. [8] pointed out that the main factor affecting the storage capacity during the construction of a UGS is the gas injection rate. Ban et al. [9] found that reservoir physical properties, water invasion and gas and water seepage variation are main factors affecting the efficiency

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of UGS construction, and pointed out that there is a certain correlation between the injection-production rate and gas-flooding efficiency. Du et al.^[10] conducted a sensitivity experiment on gas injection and production rates, and identified the relationship between gas injection rate and storage capacity, gas production rate and efficiency for gas-driven water when constructing a reservoir into a UGS. However, due to the small core size used in the experiment, it was difficult to clearly understand the fluid distribution and expansion law during the operation of the UGS.

At present, the seepage mechanism of reconstructing a water-drive gas reservoir into a UGS is mainly carried out by physical simulation using conventional cylindrical cores (2.5 cm or 3.8 cm in diameter, and 2 to 3 times the diameter in length) or by two-dimensional and three-dimensional numerical simulation methods to study the pore producing law and expansion mechanism of the reservoir. Under the influences of single seepage dimension, small core size and small total pore space, the simulation accuracy is low and the measurement error is large. The experiment results are very limited, and difficult to reflect the pore producing law in the two-dimensional seepage process in real formation^[11–14]. Flat-plate models are mainly used to simulate reservoir development, but they have not been used in injection-production experiments on UGSs.

In this paper, an experimental method for simulating UGS injection and production is established based on an independently developed large-scale two-dimensional flat-plate high-pressure system, in order to investigate the production dynamics and pressure changes during UGS construction, multi-cycle injection-production process and identify the factors affecting storage capacity and expansion under varying permeability conditions.

1. Experiment design

All one-dimensional cylindrical cores and two-dimensional flat cores are made of the same batch of outcrop rock samples to perform gas injection storage construction and operation tests using the one-dimensional cylindrical core and two-dimensional flat core models. A total of two cylindrical cores and two flat cores were made for testing. Cores of the same type are similar in size. The cylindrical core is 3.8 cm in diameter and 30.1 cm long while the size of flat core is 30 cm × 30 cm × 4.5 cm. The absolute permeability of 1[#] cylindrical core and 1[#] two-dimensional flat core is $104.0 \times 10^{-3} \mu\text{m}^2$, and that of 2[#] cylindrical core and 2[#] flat core is $1.4 \times 10^{-3} \mu\text{m}^2$.

1.1. One-dimensional cylindrical core experiment

The experiment on the one-dimensional cylindrical cores was conducted to investigate the gas injection and production volumes and gas loss variation law during UGS operation under different water saturations. The

result provided a basis for dividing the types of working zones in subsequent two-dimensional flat core experiments. During the experiment, two cylindrical cores and 99.99% pure CH₄ were used to perform injection and production under varying water saturation conditions. Fig. 1 shows the experimental setup. The experiment procedures are as follows: (1) Vacuumize 1[#] cylindrical core and create different water saturation environments (50%–98%) using the steady-state method. To ensure consistent experiments, pressure and differential pressure settings during the one-dimensional core experiment were consistent with the two-dimensional flat core plate experiment, and set the system back pressure at 8 MPa; (2) Close the outlet, and enhance the pressure to 16 MPa by injecting gas at a constant speed (1.0 mL/min), and then kept standing for 12 h; (3) Open the outlet, and then produce gas at a constant speed (1.5 mL/min). Record the gas volume and the fluid volume produced, and end the experiment until no fluid from the outlet; (4) Use 2[#] cylindrical core and repeat steps (1)–(3).

1.2. Two-dimensional flat core experiment

For the preparation of two-dimensional flat core models: 17 well sites were located on a flat core for pressure monitoring as well as injection and production, and 32 pairs of electrodes were placed on the core to monitor the variation of gas and water migration and overall water saturation during UGS construction and operation. Finally, the whole model was cast with epoxy resin to form an isolation layer on the core surface (Fig. 2).

The high-temperature and high-pressure experiment setup (Fig. 3) consists of a core model, a high-temperature and high-pressure chamber, a multi-point saturation/pressure acquisition system, a temperature acquisition and control system, a flow control valve, a high-pressure drying unit, gas-water separator, a back pressure pump and a confining pressure pump. Using the saturation and pressure monitoring system, the multi-point saturation and pressure of the core model could be monitored in real time while gas injection and production rates could be controlled by throttling the cross-section of

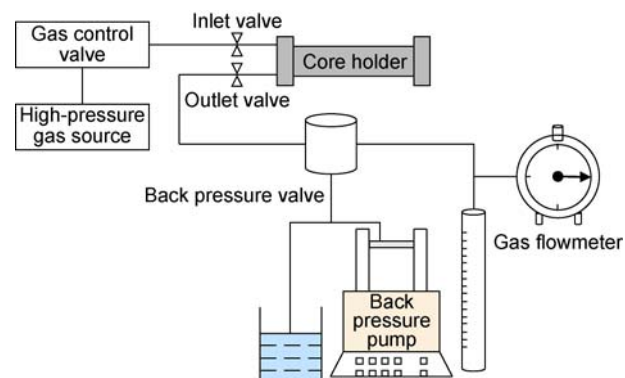


Fig. 1. Experiment setup for gas injection and production on one-dimensional cores.

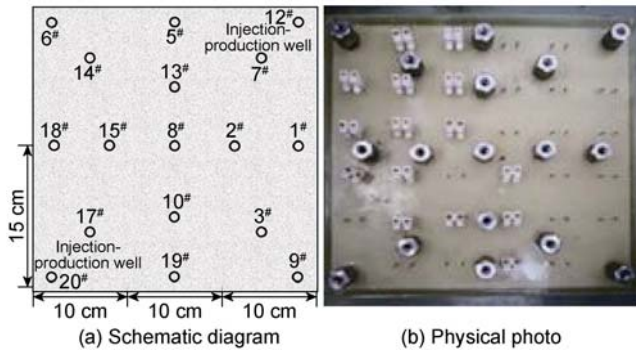


Fig. 2. Schematic diagram and physical photo of the two-dimensional flat core model.

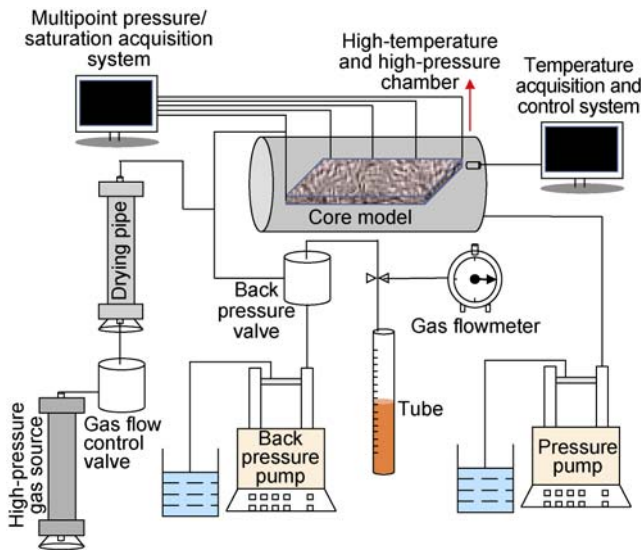


Fig. 3. Schematic diagram of the high-temperature and high-pressure experiment system for flat core model.

the gas flow controlling valve.

Injection-production parameters: Taking the Hutubi UGS in Xinjiang, NW China, as an example, the formation thickness of the UGS is 50 m, the well spacing is 100 m, and the injection and production volumes per well are $66.5 \times 10^4 \text{ m}^3/\text{d}$ and $100.0 \times 10^4 \text{ m}^3/\text{d}$ respectively. According to the theory that the average flowing speed is similar among all pores [15] and the parameters of the physical model, the UGS injection and production rates are calculated to be 0.30 mL/min and 0.45 mL/min, respectively.

The following steps were taken during the experiment:

(1) 1[#] two-dimensional flat core model was loaded into the

high-temperature and high-pressure chamber for 24 h under vacuum. Then the 20[#] well was opened to naturally absorb water until the model was saturated, and then, water was injected at constant pressure difference (0.1 MPa) through the well for 3 d. The saturated water volume and porosity were calculated according to the difference between the injected and the produced fluid volumes. (2) With the back pressure maintaining at 8 MPa, gas was injected into 20[#] well at 0.30 mL/min while fluid was produced from 12[#] well to displace water with gas until no more water was produced from the well. Then, the well was closed, and the overall produced water volume was calculated. (3) Gas was injected into 20[#] well at 0.30 mL/min until the overall pressure of the model reached 16 MPa (the peak pressure of UGS). The first cycle of storing gas was ended. (4) With the back pressure controlled at 8 MPa, 20[#] well was opened for gas production at a constant rate (0.45 mL/min). Water saturation was monitored at all test points in real time, and the produced fluid and gas volumes were calculated until no more fluid was produced from the well. (5) Gas was injected into 20[#] well at a constant rate (0.30 mL/min), and the inlet pressure of the model was monitored in real time until it increased to the upper limit (16 MPa). Then the well was closed for 12 h to balance the pressure inside the model. (6) Repeated steps (4) and (5) to complete 6 cycles of injection and production. (7) 2[#] two-dimensional flat core model was used for test and repeated steps (1)–(6).

2. UGS building and operating dynamics of one-dimensional cylindrical core model

Fig. 4 presents the variation of gas injection and production capacity and gas loss rate (the ratio of the difference between injection volume and production volume to injection volume during the injection and production) under different water saturation levels in cylindrical cores. The results indicate that the injection and production capacity of the model is negatively correlated with water saturation. In other words, better physical properties resulted in stronger injection and production capacity (Fig. 4a, 4b) and higher gas injection and production volumes. During gas injection storage building, higher

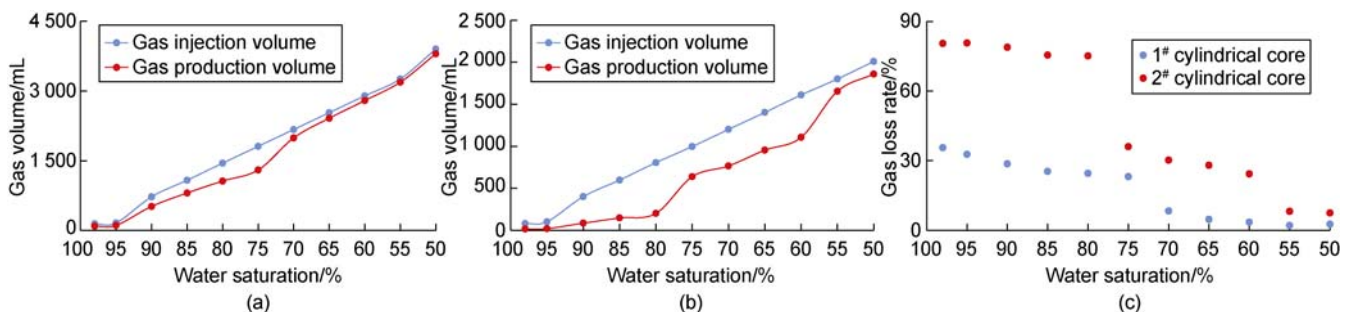


Fig. 4. Injection-production volume dynamics of cylindrical core model. (a) Volume of gas injection and production of 1[#] cylindrical core; (b) Volume of gas injection and production of 2[#] cylindrical core; (c) Gas loss rate.

water saturation led to a higher gas loss rate (Fig. 4c). When the physical properties of the core were improved, the gas loss rate showed "two-stage" distribution characteristic. The gas loss rate was higher (23%–36%) when the water saturation was greater than or equal to 75%, but it significantly decreased (below 10%) when the water saturation was less than 75%. On the other hand, when the physical properties were worse, the gas loss rate was distributed in "three-stages". The loss rate was at a very high level (above 75%) when the water saturation was greater than or equal to 80%, then declined but remained high (31%–36%) when the water saturation ranged from 60% to 75%, and finally dropped dramatically to less than 10% once the water saturation was less than or equal to 55%.

According to the injection-production volume dynamics of the one-dimensional core model, water saturation is a key factor affecting gas loss in the UGS. If there is a local flooded layer in the storage space of the UGS, the injected gas would be lost greatly in this high water-cut zone, which may increase the ineffective storage capacity and gas loss of the UGS. The problem can be addressed by reducing water saturation by taking measures such as displacement drainage.

Notably, the permeability of 1[#] cylindrical core is higher. The gas loss rate was higher and the gas flow in the core became more difficult when the water saturation was greater than 75%, and the gas loss rate greatly reduced as water saturation dropped below 75%. Accordingly, the working zones of the model can be divided into two types: a high-efficient working zone characterized by water saturation less than 75%, and a low-efficient working zone where water saturation is greater than or equal to 75%. Similarly, with the low permeability of the 2[#] cylindrical core, the working zones can be grouped into three types: an ineffective working zone with water saturation greater than 80%, a low-efficient working zone with water saturation of 55%–80%, and a high-efficient working zone with water saturation less than 55%.

3. 2D flat core gas storage model construction and operating dynamics

3.1. UGS construction through water displacement

Fig. 5 shows the curves of injection pressure and the gas-liquid ratio during water displacement and drainage from the flat core model. The injection pressure rose at the beginning of the UGS construction, then the injection pressure started to drop when gas was observed at the outlet. Compared with the increase in the gas-liquid ratio, there was a hysteresis in the pressure drop, which is related to the pressure transfer. When a large amount of gas was observed at the outlet (the gas-liquid ratio greater than 500 m³/m³), the injection pressure leveled off, followed by a rapid rise in the gas-liquid ratio, and the

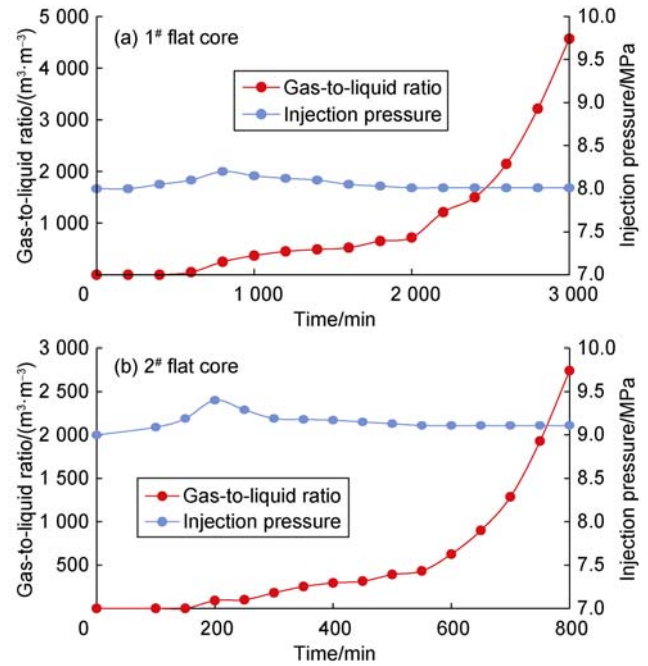


Fig. 5. Dynamic curves during water displacement in the flat core model.

produced fluid was mainly in gas phase when the storage construction was completed.

Fig. 6 displays the distribution of water saturation within the model at different time points during water displacement and drainage from the flat core model. Comparative analysis indicates that gas-flooding efficiency improved when the permeability of the flat core increased. The gas swept zone in 1[#] flat core was much larger than that in 2[#] flat core, whether before, when or after gas broke through the outlet. This is attributed to lower permeability and greater flowing resistance in the core, which more likely creates preferential channels and a smaller sweep zone.

3.2. Multi-cycle injection-production

When the flat core model moved into the stage of multi-cycle injection-production, the formation water was continuously produced after gas was injected, and the capacity of the UGS model became larger. Fig. 7 illustrates the water saturation distribution at the end stage of production in 1[#] flat core model after different cycles of gas injection-production. After four cycles, the effective working zone in the UGS model was mostly distributed near the injection well, and gradually expanded to the both sides of the preferential gas channels formed after water was displaced, and the gas saturation in the effective working zone increased (Fig. 7a, 7b). The gas swept zone and the gas saturation in the swept zone did not change anymore from the fifth to sixth cycles (Fig. 7c, 7d).

Fig. 8 shows the relationship between the volumes and cycles of gas injection-production in 1[#] flat core model. There was a large difference between the injection volume and the production volume at the end of the first

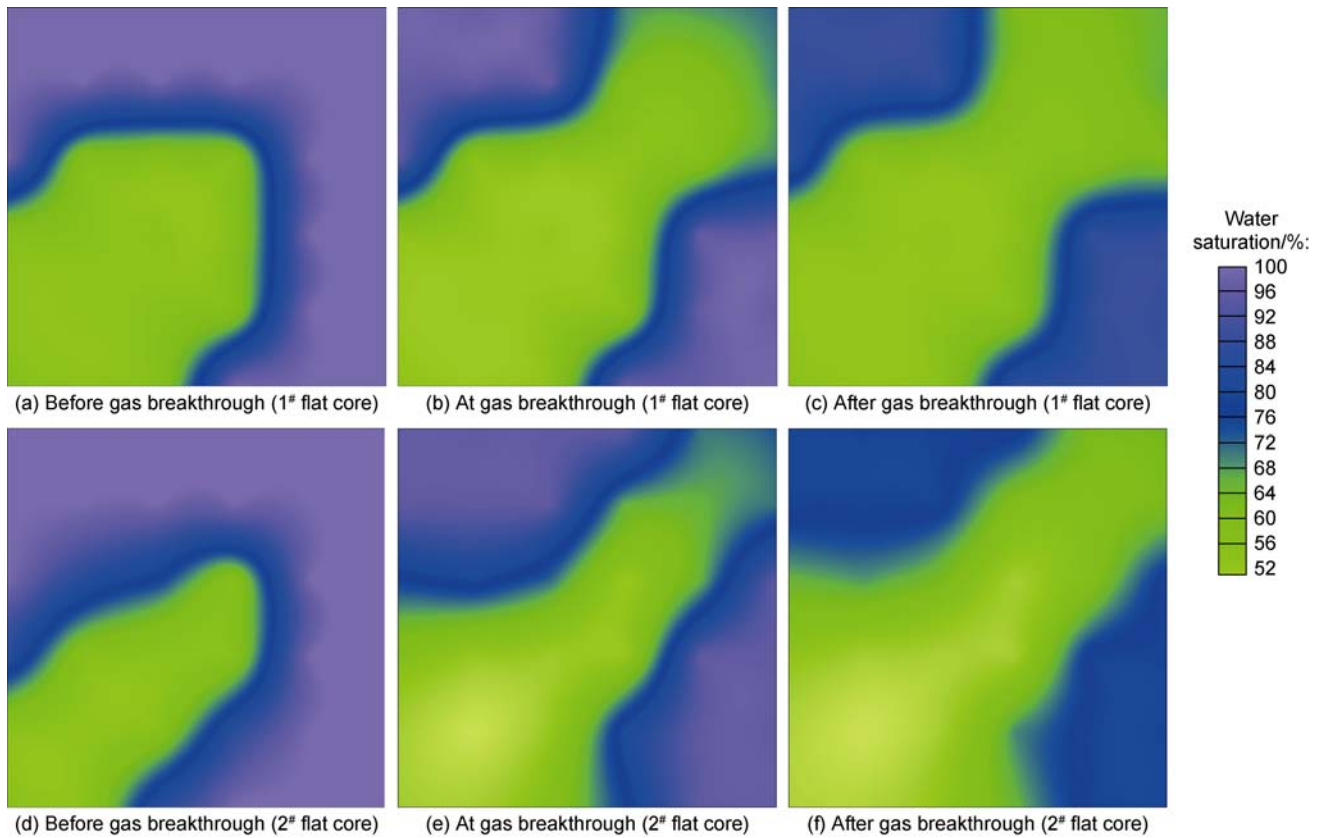


Fig. 6. Distribution of water saturation at different time points during water displacement from the flat core model.

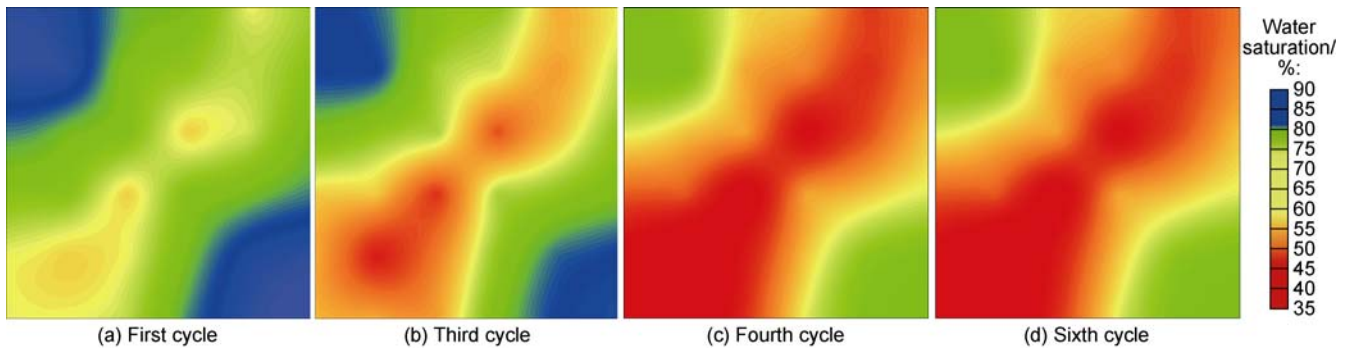


Fig. 7. Distribution of water saturation at the end of production stage in 1# flat core model after different cycles of gas injection-production.

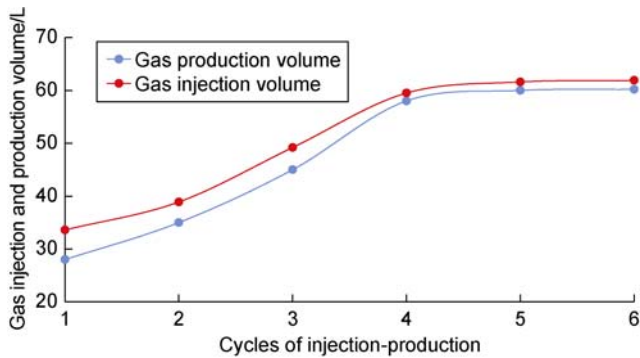


Fig. 8. Curves of gas injection and production volumes of 1# flat core model.

four cycles, which showed an increasing trend. Although there was still a difference between the injection volume and the production volume after the fifth and sixth cycles,

the difference reduced greatly and became stable, and the curves changed from an increasing trend to a horizontal trend. Overall, the maximum storage capacity of 61.6 L was reached after four cycles of injection-production in 1# flat core model. The efficiency of formation water displacement (the percentage of produced water to the saturated water of the model) was calculated to be 26.5% after the maximum capacity was reached.

The variation of the storage capacity of the 2# flat core model is similar to that of the 1# flat core model. The effective working zone of the UGS model expanded to both sides of the preferential gas channels in the first four cycles of injection-production, and the gas swept zone and the gas saturation in the swept zone did not change much during the last two cycles (Fig. 9). However, the model expansion was slow due to the low permeability of

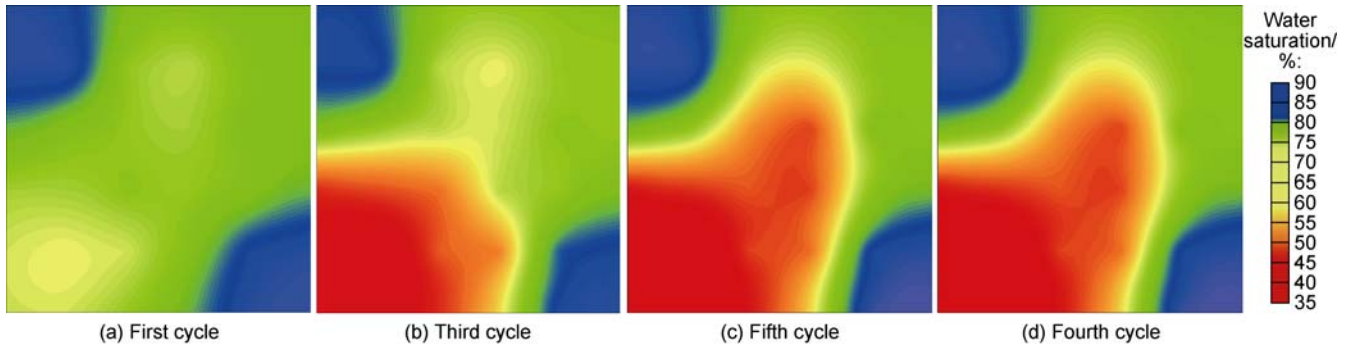


Fig. 9. Distribution of water saturation after different cycles of gas injection-production in 2[#] flat core model.

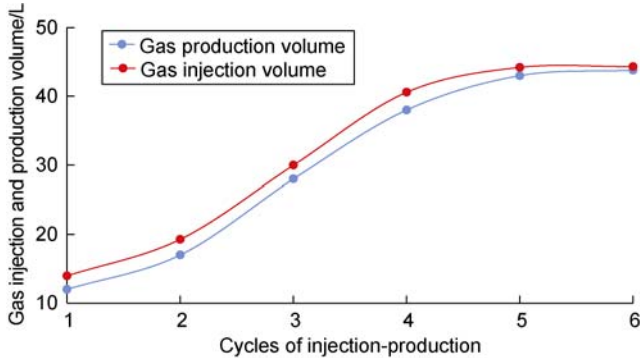


Fig. 10. Curves of gas injection and production volumes of 2[#] flat core model.

the core. The maximum storage capacity of 44.3 L was finally reached after five cycles of injection-production (Fig. 10). The efficiency of water displacement after reaching the maximum capacity was calculated to be merely 14.7%, far lower than that of the 1[#] flat core model.

The working zones of the flat core model at the maximum capacity were classified according to the criteria for the cylindrical core model (Fig. 11). (1) In 1[#] flat core model, the high-efficient working zone accounts for about 73% of the total area. Since the gas saturation in this zone was high, there was basically no free water, and the gas flow resistance during the UGS operation stage was low. The low-efficient working zone where some free water existed covered approximately 27.0%, and free water affected the gas flow during UGS operation. There was no inefficient working zone in this model. (2) The high-efficient working zone accounts for 45.0% of the total area in

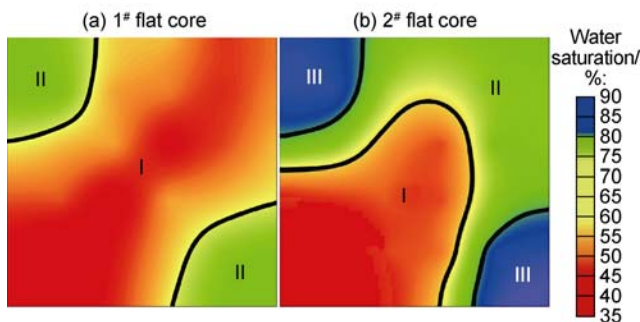


Fig. 11. Work zones of the flat core models at the maximum capacity. I—High-efficient working zone; II—Low-efficient working zone; III—Inefficient working zone.

the 2[#] flat core model, the low-efficient working zone accounts for about 44.6%, and the ineffective working zone accounts for 10.4%. The better reservoir physical properties of the UGS bring a larger high-efficient working zone.

3.3. Factors affecting the storage capacity of the UGS model

According to the experimental results, there are three factors affecting the storage capacity during the operation of the UGS model. (1) Permeability: The higher permeability of the UGS model led to the higher efficiency of water displacement, higher utilization rate of storage space, and larger UGS capacity. In the model with higher permeability, the ineffective working zone was small or inexistent, the loss of injected gas was less, and the utilization rate was higher than those of the model with lower permeability. (2) Water saturation: The experiment on the cylindrical core model suggests that gas loss rose with the increase in water saturation. Lowering water saturation by optimizing gas injection-production and water displacement could effectively reduce gas loss in the UGS. (3) Injection-production equilibrium time: From monitoring the pressure variations during the equilibrium of multi-cycle injection-production in the flat core model (Fig. 12), the bottom hole pressure (BHP) decreased slightly in the early equilibrium phase (less than 0.5 MPa) and then became stable after 4 h. Therefore, extending the equilibrium phase did not contribute more to the

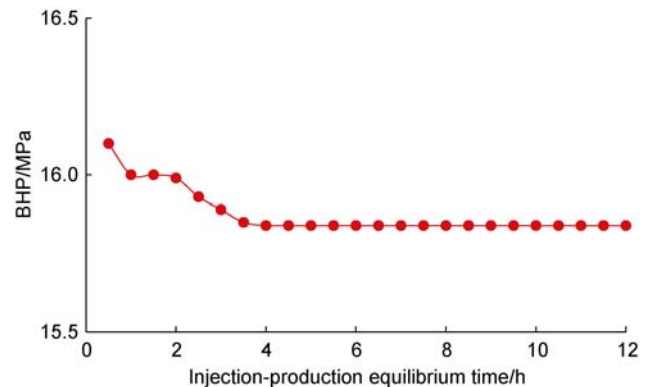


Fig. 12. Pressure variations in the first cycle of injection-production equilibrium phase in 1[#] flat core model.

expansion of the gas storage.

4. Conclusions

In the process of gas reservoir-type UGS capacity construction and operation, the storage space can be divided into three zones: high-efficient working zone, low-efficient working zone and ineffective working zone. Higher reservoir permeability leads to higher pore utilization efficiency, smaller or no ineffective working zone, less gas loss and a higher gas utilization rate.

Better reservoir physical properties and large storage space contribute to a larger UGS capacity. Higher water saturation results in more gas loss during UGS capacity construction and operation. Optimizing water displacement and lowering the water saturation may reduce gas loss.

During multi-cycle injection and production, there is a reasonable equilibrium time interval of gas injection and production. Beyond this interval, extending the equilibrium phase won't contribute more to the expansion of UGS capacity.

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